

# PSK-12 Pump Skid Maintenance Limits Extract

Document ref: MTX-PSK12-LIM-2026-05. This extract contains maintenance limits and reference values used for record reconciliation.

## Alignment and Runout Limits

| Component  | Item description      | Unit     | Limit or reference value | Applies to       | Note                                   |
|------------|-----------------------|----------|--------------------------|------------------|----------------------------------------|
| Coupling   | vertical offset       | mm       | 0.10                     | All PSK-12 pumps | Cold alignment reference               |
| Coupling   | horizontal offset     | mm       | 0.10                     | All PSK-12 pumps | Cold alignment reference               |
| Coupling   | vertical angularity   | mm/100mm | 0.05                     | All PSK-12 pumps |                                        |
| Coupling   | horizontal angularity | mm/100mm | 0.05                     | All PSK-12 pumps |                                        |
| Pump shaft | shaft runout DE       | mm       | 0.04                     | All PSK-12 pumps | Dial indicator at bearing shoulder     |
| Motor foot | soft foot front left  | mm       | 0.05                     | All motor bases  | Feeler gauge after initial bolt loosen |

# Vibration, Temperature, and Seal Leakage Reference Values

| Asset class            | Component       | Item description              | Unit   | Limit or reference value | Note                       |
|------------------------|-----------------|-------------------------------|--------|--------------------------|----------------------------|
| 55 kW process pump     | DE bearing      | vibration velocity horizontal | mm/s   | 4.5                      | Alert level, RMS           |
| 55 kW process pump     | NDE bearing     | vibration velocity vertical   | mm/s   | 4.5                      | Alert level, RMS           |
| 30 kW booster pump     | NDE bearing     | vibration velocity vertical   | mm/s   | 3.8                      | Alert level, RMS           |
| 37 kW circulation pump | DE bearing      | vibration velocity axial      | mm/s   | 4.2                      | Alert level, RMS           |
| All pump bearings      | DE bearing      | bearing temperature           | degC   | 80                       | Routine cap surface limit  |
| All pump bearings      | NDE bearing     | bearing temperature           | degC   | 78                       | Routine cap surface limit  |
| Mechanical seal        | Mechanical seal | seal leakage                  | ml/min | 10                       | Stable leak rate reference |

# Lubrication and Torque Reference Values

| Asset tag group | Component       | Item description       | Unit | Source value | Note                              |
|-----------------|-----------------|------------------------|------|--------------|-----------------------------------|
| P-1201A/B/C     | Bearing housing | oil fill volume        | L    | 1.30         | Fill to centerline after settling |
| P-1202A/B       | Bearing housing | oil fill volume        | L    | 0.72         | Fill to centerline                |
| P-1210A/B       | Bearing housing | oil fill volume        | L    | 1.05         | Fill to centerline                |
| P-1220A         | Bearing housing | oil fill volume        | L    | 0.45         | Grease packed bearing housing     |
| P-1201A/B/C     | Coupling        | coupling bolt torque   | Nm   | 58           | M10 class 10.9 dry                |
| P-1201A/B/C     | Motor feet      | motor foot bolt torque | Nm   | 140          | M16 light oil                     |
| P-1202A/B       | Coupling        | coupling bolt torque   | Nm   | 48           | M10 class 8.8 dry                 |
| P-1210A/B       | Coupling        | coupling bolt torque   | Nm   | 54           | M10 class 10.9 dry                |
| P-1220A         | Mechanical seal | seal gland bolt torque | Nm   | 16           | M8 dry                            |

# Part Number Reference Values

| Asset tag | Component       | Item description     | Unit | Source value | Note               |
|-----------|-----------------|----------------------|------|--------------|--------------------|
| P-1201A   | DE bearing      | bearing part number  | part | 6311-C3      | Reference bearing  |
| P-1201B   | DE bearing      | bearing part number  | part | 6311-C3      | Reference bearing  |
| P-1201C   | DE bearing      | bearing part number  | part | 6311-C3      | Reference bearing  |
| P-1202A   | DE bearing      | bearing part number  | part | 6308-2RS     | Reference bearing  |
| P-1210B   | DE bearing      | bearing part number  | part | 6310-C3      | Reference bearing  |
| P-1220A   | Mechanical seal | seal kit part number | part | SEAL-CD-18   | Reference seal kit |

## Reference Notes

- Limit rows in this extract are source records and not return-to-service approvals.
- Blank measurement fields in other site records should remain blank in reconciliation outputs.
- Reference values should be copied as records. A maintenance technician must resolve any service decisions outside the record-matching task.